8

MANUFACTURING SYSTEMS

8.1 INTRODUCTION

Facilities design for manufacturing systems is extremely important because of the economic dependence of the firm on manufacturing performance. Since manufacturing, as a whole, is a *value-adding* function, the efficiency of the manufacturing activities will make a major contribution to the firm's short- and long-run economic profitability.

Greater emphasis on improved quality, decreased inventories, and increased productivity encourages the design of manufacturing facilities that are integrated, flexible, and responsive. The effectiveness of the facility layout and material handling in these facilities will be influenced by a number of factors, including changes in

- Product mix and design
- Materials and processing technology
- Handling, storage, and control technology
- Production volumes, schedules, and routings
- Management philosophies

Today's environment requires that companies quickly respond to varying customer requirements. Manufacturing strategies, such as *just-in-time* (JIT) production, lean manufacturing, and others have emerged as viable and effective methods for achieving efficiency. Their successes have been rooted in the total commitment by companies to solve their problems on a daily basis.

But first, let us take a historical perspective. Not too long ago, automation was viewed as the panacea of many U.S. manufacturers. One goal then was to build and operate a totally lights-out factory—that is, to build the automatic factory.

The concept of an automatic factory has captured the imagination of both managers and engineers throughout the world. The automatic factory can be distinguished from the *automated* factory as follows: the automatic factory is essentially a paperless factory. In the automated factory, automation and mechanization are dominant; however, people perform a limited number of direct tasks and a greater number of indirect tasks. In the automated factory, factory personnel resolve unusual situations; in the automatic factory, every effort is made to ensure that unusual situations do not arise.

Not all facilities planning efforts involving new production plants will result in automatic factory designs. To the contrary, it is likely that few will justify such levels of automation, at least in the short run. However, in developing strategic facilities plans, it is important to plan for upgrading facilities to accommodate new technology, changing attitudes concerning automation, and increased understanding concerning factory automation. Firms producing high-technology products may choose to subcontract component manufacturing and labor-intensive activities. They may focus only on high-technology design and processing activities that represent significant value-added contributions.

Many external factors affect the facility planning process. Among the factors that appear to have a significant impact are

- Volume of production
- Variety of production
- Value of each product

Each of these factors may lead to different types of manufacturing facilities (e.g., job shop, production line, cellular manufacturing, etc.). In complex facilities, all types of manufacturing systems will most likely exist.

Facilities development often involves an incremental approach to making the transition from a conventional or mechanized factory to an automated factory. While this strategy may be the only viable alternative due to limited capital investment dollars, the ultimate goal should be to bridge the pockets of automation and move toward an integrated factory system. Integration does not necessarily mean automation. In some situations, “information integration” may be the only choice with the physical system not integrated. This situation is often observed in older facilities.

From a system viewpoint, an incremental approach is not necessarily bad, so long as the steps are considered to be interim steps in a phased implementation of an automation project. However, to obtain an integrated factory system, the subsystems must be linked together. An obvious approach is to build physical bridges that join together automated subsystems. Likewise, information bridges can be provided through a factory-wide control system.

The modern factory must have a brain and circulation system; plant-wide control system software must tie all the automated hardware subsystems together with an integrated, automated material handling system. The software requirements include the following:

- Integrate material handling information flows with shop floor control information.
- Assign and schedule material handling resources.
- Provide real-time control of material move, store, and retrieve actions.

The ability to know when and where the resources are needed is necessary to achieve proper control. Detailed information on every product and major components found in the bill of materials, as well as tools, fixtures, and other production resources, must be captured and maintained in real time. The implementation of an automated data capture and communication system is a critical element to achieving a high-performance manufacturing system.

The critical types of information needed to integrate a plant-wide material handling system include

- Identification and quantification of items flowing through the system
- Location of each item
- Current time relative to some master production schedule

Such information is necessary in order to synchronize the multitude of transactions that take place in a manufacturing system.

Alternative routing paths and buffer storage are two methods that may protect the factory from catastrophic interruptions and delays. However, to make use of alternative paths, a real-time dispatching decision must be made. The material handling system control must have the flexibility to determine which of several alternate paths should be taken and must have the physical resources needed to execute the decision.

In designing an integrated material handling system, the following objectives should be considered:

- Create an environment that results in the production of high-quality products.
- Provide planned and orderly flows of material, equipment, people, and information.
- Design a layout and material handling system that can be easily adapted to changes in product mix and production volumes.
- Reduce work in process and provide controlled flow and storage of materials.
- Reduce material handling at and between workstations.
- Deliver parts to workstations in the right quantities and physically positioned to allow automatic transfer and automatic parts feeding to machines.
- Deliver tooling to machines in the right position to allow automatic unloading and automatic tool change.
- Utilize space most effectively, considering overhead space and impediments to cross traffic.

But more importantly, products must be designed for both manufacturability and ease of handling. Specifically, the shapes and sizes of materials, parts, tooling, subassemblies, and assemblies must be carefully considered to ensure that automatic transfers, loading, and unloading can be performed.

The past decades have been characterized as the era that strived to develop the automatic factory: a noble goal, perhaps, but looking back it seems to have been too ambitious. Many of today's factories do not strive to be fully automated. The current thinking is more toward finding the right blend of machine and human performance characteristics. The primary goal is to increase customer satisfaction at a reasonable price. Achieving the goal requires that the company reduce work in process, increase performance of individual machines, reduce capital cost at the same level of production capacity, and achieve higher productivity from factory personnel.

We now look at several types of manufacturing systems. They are representative of those commonly found in industry.

8.2 FIXED AUTOMATION SYSTEMS

8.2.1 Transfer Line

In a transfer line, materials flow from one workstation to the next in a sequential manner. Because of the serial dependency of the transfer line, the production rate for the line is governed by the slowest operation. A transfer line is one example of hard automation.

Transfer lines are often used for high-volume production and are highly automated. In highly automated lines, the processing rates of individual machines are matched so that there is usually no need for buffer storage between machines. Or, if there is buffer storage, it would be for reasons of unexpected machine breakdowns.

The transfer line offers production rates unmatched by other types of manufacturing systems. But its disadvantages are

- Very high equipment cost
- Inflexibility in the number of products manufactured
- Inflexibility in layout
- Large deviation in production rates in case of equipment failure in the line

Many features of a transfer line can be observed on manual, paced assembly lines. Inventory banks or buffers can be placed between workstations to compensate for variations in production rates at individual workstations. Production rate variations occur due to machine failures, the inherent variation in operator performance, uncertainty in the arrival of components needed at individual workstations, and other causes.

The design of transfer lines includes both the specification of the individual processing stages and the linkage of the stages. The performance of the system is dependent on the layout of the facility, the scheduling of production, the reliability of the individual stages in terms of processing variability and machine failures, and the loading of the line. Facility planning for the transfer line is relatively straightforward. The processing equipment is arranged according to the processing sequence. Buffer sizes between workstations must be determined and accommodated by several types of material handling devices such as vertical storage systems or spiral-type conveyors. Or the spacing between machines can be increased to accommodate buffer stock.

As a result of the predictable flow sequences, straight-line flow and its derivatives are frequently used. Figure 8.1 shows several variations of the straight-line flow structure.

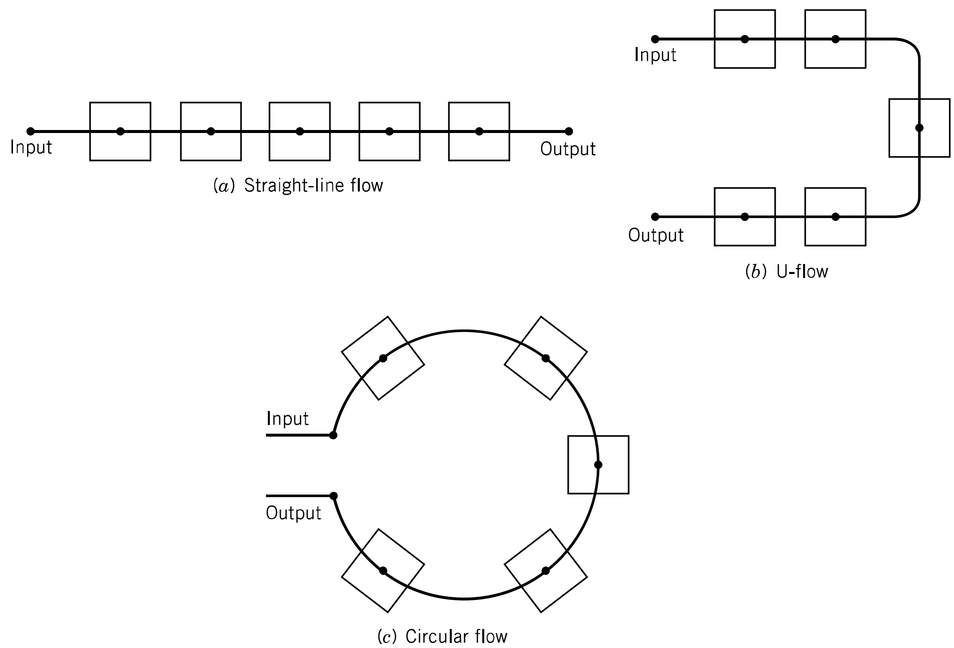


Figure 8.1 Several variations of the straight-line flow pattern.

8.2.2 Dial Indexing Machine

One particular implementation of the circular flow is the dial indexing machine [10], where the workstations and the input/output stations are arranged in a circular pattern. The worktable where the parts are mounted is indexed clockwise or counterclockwise at predetermined times based on the required processing rate. This configuration corresponds to the circular pattern in Figure 8.1*c*.

Although they are not referred to as transfer lines, a number of production systems have characteristics that are similar to those of transfer lines. A few examples include

- Automated assembly systems for automotive parts
- Chemical process production systems
- Beverage bottling and canning processes
- Heat treatment and surface treatment processes
- Steel fabrication processes

The common characteristic among the above examples is that the automation is fixed and the processing is synchronous. Parts go from one machine to another without manual intervention. Materials are transferred from one machine to the next in sequence by an automated material handling system.

As with transfer lines, the development of the layout and material handling system involves the placement of workstations in processing sequence, the specification of spaces between workstations, and the determination of the type of handling method to use.

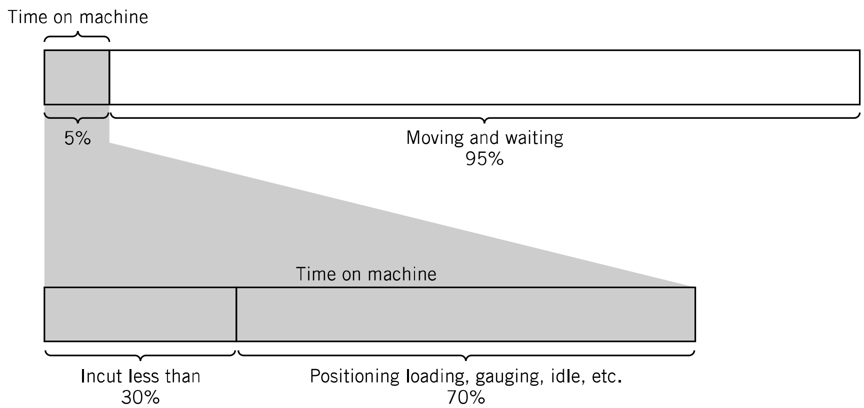


Figure 8.2 Pictorial representation of work in process after being released to the shop floor. (Merchant [18].)

8.3 FLEXIBLE MANUFACTURING SYSTEMS

In a study of batch-type metal-cutting production, Merchant [18] reported that the average workpiece spends only about 5% of its time on machine tools (see Figure 8.2).

Furthermore, of the time the material is loaded on a machine, it is being processed less than 30% of the time. It is argued that gaining control of the "95% of the time parts are not machined" involves linking the machines by an automated material handling system and computerizing the entire system and operation. The term *flexible manufacturing system* (FMS) has thus emerged to describe the system.

The word "flexible" is associated with such a system since it is able to manufacture a large number of different part types. The components of a flexible manufacturing system are the processing equipment, the material handling equipment, and the computer control equipment. The computer control equipment is used to track parts and manage the overall flexible manufacturing system.

Groover [10] lists the following manufacturing situations for which a flexible manufacturing system might be appropriate:

- Production of families of work parts
- Random launching of work parts onto the system
- Reduced manufacturing lead time
- Increased machine utilization
- Reduced direct and indirect labor
- Better management control

In designing the material handling system for a flexible manufacturing system, the following design requirements are recommended by Groover [11]:

- Random, independent movement of palletized parts between workstations
- Temporary storage or banking of parts
- Convenient access for loading and unloading

- Compatibility with computer control
- Provision for future expansion
- Adherence to all applicable industrial codes
- Access to machine tools
- Operation in shop environment

Flexible manufacturing systems are used to cope with change. Specifically, the following changes are accommodated:

- Processing technology
- Processing sequence
- Production volumes
- Product sizes
- Product mixes

Flexible manufacturing systems are designed for small-batch (low-volume) and high-variety conditions. Whereas hard automation manufacturing systems are frequently justified on the basis of economies of scale, flexible automation is justified on the basis of scope. The concept of flexible manufacturing systems is one that has very broad scope.

Flexibility can be accomplished through

- Standardized handling and storage components
- Independent production units (manufacturing, assembly, inspection, etc.)
- Flexible material-delivery system
- Centralized work-in-process storage
- High degree of control

Because of the variety of alternatives for storing, handling, and controlling material, the specification of the material handling system and design of the layout can be quite varied. Figure 8.3*a* and *b* show two alternative configurations for the same flexible manufacturing system. Figure 8.3*a* illustrates an external centralized work-in-process storage, while Figure 8.3*b* shows an internal centralized work-in-process storage. Work in process is needed because a part has to go through several machines before it eventually leaves the system after the last machining operation is performed. A more detailed discussion on reduction of work in process is provided in Section 8.5.

Yet another FMS configuration based on cellular manufacturing principles is illustrated in Figure 8.3*c*. In this configuration, the handling distances are reduced significantly as the machines are placed within the work envelope of the transfer device, a robot handler in this illustration. This configuration has some built-in scalability as only one cell can be used in case there is a reduction in the demand for the products that are machined in this system.

We conclude from these illustrations that there are multiple alternative configurations to the FMS cell design problem. The determination of which configuration is best requires a detailed comparison using analytical tools.

What makes FMS flexible? According to Groover [10], for a manufacturing system to be categorized as flexible, it must have the capability to do the following:

1. Process different part styles in a non-batch mode.

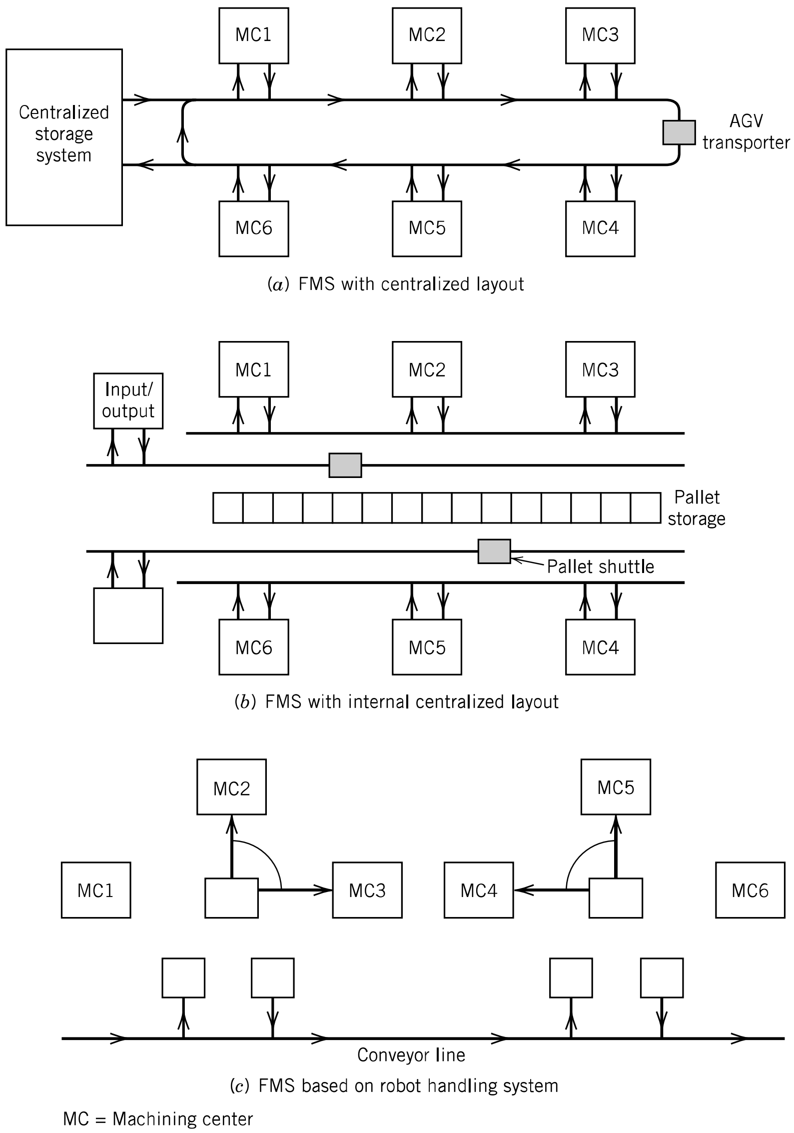


Figure 8.3 Alternative layouts for a flexible manufacturing system.

2. Accept changes in production schedule.
3. Respond gracefully to equipment malfunction and breakdowns in the system.
4. Accommodate the introduction of new part designs.

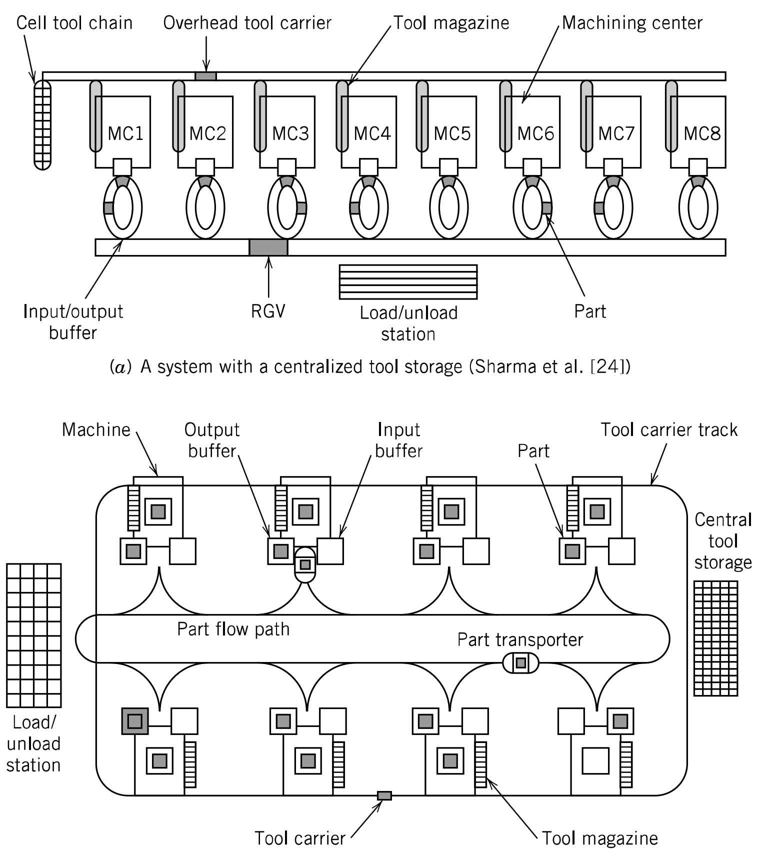
From these conditions, we can say that an automated system is not always flexible (e.g., transfer lines), and vice versa, that a flexible system need not be automated (manual assembly lines).

In the next section, we consider a special case of flexible manufacturing system, the single-stage multimachine system (SSMS).

8.4 SINGLE-STAGE MULTIMACHINE SYSTEMS

Yet another alternative in automation in machining systems is what has been termed *single-stage multimachine systems* (SSMS). The discussion below is based on Koo [12] and Koo and Tanchoco [13]. SSMS is described in terms of the resources involved.

1. *Manufacturing configuration and machines.* The machining centers in SSMS are very versatile machines. They are all identical and versatile so that all operations on any part can be performed on any one machine. This being the case, there will be only one setup per part. There is a tool magazine on each machine, but with limited capacity. The tool-handling system supplies the tools that are not resident on the machine. Each machine has a separate input and output spur with limited capacity. Once a part is loaded on a machine, it does not leave the machine until all required operations are performed. Thus, the part is moved to and from the machine only twice. Under this type of manufacturing system, the tool-delivery system becomes the critical resource. Figure 8.4 illustrates two alternative tool-delivery systems in a single-stage multimachine system.



(*a*) A system with a centralized tool storage (Sharma et al. [24])

(*b*) A system with distributed tool storage (Koo [12,13])

Figure 8.4 Alternative tool-handling system in single-stage multimachine systems.

2. *Parts.* Parts arrive according to the production requirements schedule. As mentioned above, parts visit only one machine since all operations can be performed in a single machine. A part transporter system handles the delivery of parts from the input station to the machine and from the machine to the output station. A part cannot be processed unless the required tool is available.
3. *Tools.* The required tools are specified by the process plan. Some tools are resident to the machine, while others are stored in a centralized tool storage system. These tools are generally expensive, and storing a complete tool set resident in a machine will be very expensive. Dynamic sharing of tools may be the economic choice. Still, there is a decision that must be made on what percentage of the tools should be resident or shared.
4. *Part transporter.* The part transportation activity is minimal since a part visits a machine only once. Deadheading still occurs, and there must be an efficient dispatching algorithm for the vehicles. Vehicles are dispatched based on part arrivals and job completions when vehicles are idle or when vehicles become idle with one or more parts waiting to be transported.
5. *Tool carriers.* A tool carrier handles the transport of tools to and from the centralized tool storage and the machines requiring the tool. Under dynamic tool sharing, these tools are dynamically dispatched to the machines needing them. One can also reallocate the tools when opportunities arise. Figure 8.5 illustrates a combined schedule of machines and tools. The determination of the combined schedule of machines and tools is a critical element in the operation of SSMS.

The same layout can be used for both FMS and SSMS; the difference is in the composition of the machine tools. In SSMS, the machine tools are identical, versatile machines. Transfer lines, flexible manufacturing systems, and single-stage multimachine systems address only one aspect of manufacturing: the machining operations. Furthermore, they are all automated systems, typically operating without manual intervention. What about manufacturing systems with people and machines working together? Is it possible to attain the efficiency of transfer lines with the flexibility of the flexible manufacturing system? But first we will look at the work in process and discuss how control of work in process may be accomplished. The topic of Section 8.5 is reduction of work in process.

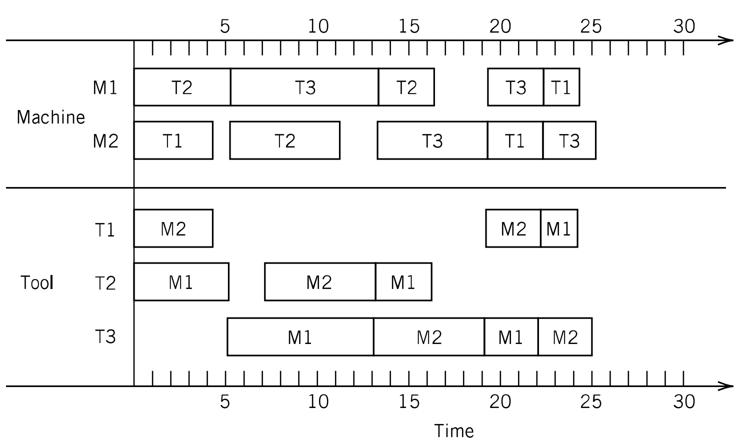


Figure 8.5 A combined schedule of machines and tools. (Koo [12, 13].)

8.5 REDUCTION OF WORK IN PROCESS

In this section, we focus on issues related to handling and storing operations in a manufacturing environment. In-process handling includes the movement of material, tooling, and supplies to and from production units, as well as the handling that occurs at a workstation or machine center. The term *in-process storage* can include the storage of material, tooling, and supplies needed to support production. However, as shown in Figure 8.6, the term normally applies to the storage of material in a semi-finished state of production.

Several rules of thumb can be used in designing in-process handling and storage systems. Among them are the following:

- Handling less is best.
- Grab, hold, and don't turn loose.
- Eliminate, combine, and simplify.
- Moving and storing material incur costs.
- Pre-position material.

Handling less is best suggests that handling should be eliminated if possible. It also suggests that the number of times materials are picked up and put down and the distances materials are moved should be reduced.

Grab, hold, and don't turn loose emphasizes the importance of maintaining physical control of material. Too often parts are processed and dumped into tote boxes or wire baskets. Subsequently, someone must handle each part individually to orient and position it for the next operation.

Eliminate, combine, and simplify suggests the principles of work simplification and methods improvement are appropriate in designing in-process handling and storage systems. Handling and storage can frequently be completely eliminated by making changes in processing sequence or production scheduling. Certainly, it is possible to combine handling tasks through the use of standardized containers.

Moving and storing material incur costs serves as a reminder that inventory levels should be kept as small as possible. Reducing inventories is one of the goals of JIT production and lean manufacturing. The underlying principle is to move material only when it is needed and store it only if you have to. Moving materials incurs personnel and equipment time and costs, and it increases the likelihood of product damage. Finally, moving materials requires a corridor of space for movement, and there are costs associated with building and maintaining aisle space. Basically, the longer material stays in the plant, the more costly the product will be, since no value is added to the product while material is moved and/or stored.

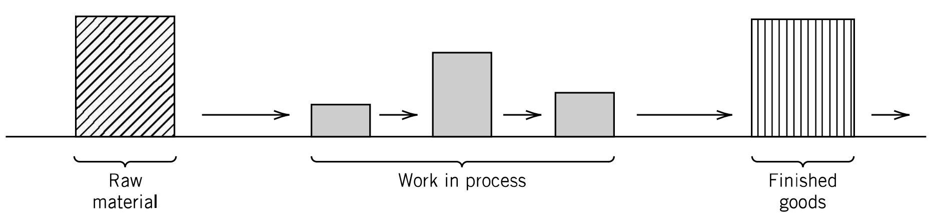


Figure 8.6 Inventories in the manufacturing cycle (Apple and Strahan [2].)

Pre-positioning of material has two aspects to be considered. First, parts should be pre-positioned to facilitate automatic load/unload, insertion, inspection, and so on. Second, when material is delivered to a workstation and/or machine center, it should be placed in a prespecified location with a designated orientation. Too often, direct labor personnel need to spend non-value-adding time to pre-position the material for machine loading.

The rules of thumb discussed above can be used to improve operations so that non-value-adding time is reduced, and consequently, the manufacturing cycle time is shortened. Reduction in wasted time increases available machine time. Reducing cycle time means that products are completed early; thus, the company can receive revenue dollars earlier. The turnover of capital is increased, resulting in lower capital costs and higher profits to the company.

Control of work in process has been the focus of many manufacturing firms. Demand-driven manufacturing systems (produce only what the customer needs at the time it is needed) have gained popularity, spawned by the success of just-in-time production at Toyota Motor Company in Japan.

8.6 JUST-IN-TIME MANUFACTURING

The *just-in-time* (JIT) production system was developed more than four decades ago by Ohno Taiichi at the Toyota Motor Company in Japan. In a broad sense, JIT applies to all forms of manufacturing, such as job shop, process, and repetitive manufacturing.

According to Ohno [22],

> No matter how difficult it may be, take it up as a challenge to reduce man-hours as a means of improving efficiency. . . . At Toyota, in order to proceed with our man-hour reduction activities, we divide waste into the following seven categories:
>
> 1. Waste arising from overproduction
> 2. Waste arising from time on hand (waiting)
> 3. Waste arising from transporting
> 4. Waste arising from processing itself
> 5. Waste arising from unnecessary stock on hand
> 6. Waste arising from unnecessary motion
> 7. Waste arising from producing defective goods

Waste arising from overproduction is a result of a production philosophy based on achieving economies of scale; that is, as the production lot increases, the unit production cost will decrease. The classic economic order quantity lot-sizing model is an example. To eliminate waste due to overproduction, Ohno states that “production lines must be reorganized, rules must be established to prevent overproduction, and restraints against overproduction must become a built-in feature of any equipment within the workplace.”

Waste from time on hand (waiting) arises when a worker tends only one machine. Assigning more machines per worker will reduce this waste. A better layout may also contribute to maximizing the number of machines that one worker can tend. And if the machines are not identical, the worker must be trained to operate the different machines involved. This has resulted in workers with multifunctional skills.

Waste from transporting comes from moving items over long distances, from work-in-process storage, and from arranging and/or rearranging parts in containers and/or pallets. The principle of simplification can be beneficial in reducing transportation waste. Waste from transporting can be eliminated if the machines are placed close to each other. By doing so, a worker can handle the transportation of parts directly from one machine to the next. The transport task can be accomplished within the machine cycle time. It is also considered a waste for parts to go through several logistic steps from the warehouse to the factory and eventually to the hands of the worker.

Waste from processing occurs when an operator is unnecessarily tied up on a machine due to poor workplace design. For example, a pneumatic clamping device to hold work parts allows an operator to do more productive work.

Reducing the labor content of the cost of making products is key to increasing the productivity of the plant. This statement does not mean that we need to automate the process; it means improving the process so that unnecessary labor time is eliminated.

We conclude this discussion by stating that the goal of JIT production is consistent with the definition of material handling given in this book. Standard and Davis [32] state that "Just-in-time means having the right part at the right place in the right amount at the right time." As you will recall, we defined material handling as "providing the right amount of the right material, in the right condition, at the right place, at the right time, in the right position, in the right sequence, and for the right costs, by using the right method."

We now look at specific ways the JIT philosophy is implemented. We categorize these methods into five elements: visibility, simplicity, flexibility, standardization, and organization.

1. *Visibility* can be obtained by the following technique: electronic boards for quick feedback, pull systems with *kanbans*,[1] problem boards, colored standard containers, decentralized storage systems, marked dedicated areas for inventory and tools, etc.
2. *Simplicity* can be achieved with a pull system with kanbans, simple setup changes, certified processes, small lot sizes, leveled production, simple machines, simple material handling, multifunctional workers, teamwork, etc.
3. *Flexibility* can be achieved with short setup times, short production times, small lot sizes, kanban carts, flexible material handling equipment, multifunctional employees, mixed-model sequencing lines, etc.
4. *Standardization* can be achieved with standard tools, equipment, pallets, methods, containers, boxes, materials, and processes.
5. *Organization* is required for setups, for cleanliness, for work areas, for the kanban system, for the storage areas, for tools, for teamwork activities, etc.

[1]A *kanban* is a card or any signal used to request or authorize production of parts; it contains information on the part, the processes used, identification of the storage area for the part, and the number of parts to produce.

At the core of the JIT philosophy is the elimination of waste. To understand waste, it is important to recognize it first. Waste can be defined as any resource that adds cost but does not add value to the product. The following are the most common sources of waste in a manufacturing facility:

- Equipment
- Inventories
- Space
- Time
- Labor
- Handling
- Transportation
- Paperwork

Equipment can be underutilized for many reasons. Some reasons are poor scheduling methods, inadequate maintenance programs, absence of feedback mechanisms to report maintenance problems, inadequate spare parts inventory, operators not trained in the use of equipment and in basic preventive maintenance, inefficient product design, long setup times and inefficient setup procedures, and large lot sizes.

Inventories for raw material, components, parts, work in progress, and finished products are usually wasted for the following reasons: large purchasing and production lot sizes, poor facility layout, inefficient material handling systems, inefficient packaging systems, inadequate training for the workers, poor organization, too many suppliers, centralized warehouses, inefficient product design, lack of standardization, awkward storage and retrieval systems, and poor inventory control systems.

Space is another key resource where waste can be identified by excess inventory, inappropriate facility layout, inadequate building design, unnecessary material handling, inefficient storage systems, and inefficient product design.

Time can be wasted due to excessive waiting or delays due to inefficient scheduling, machine downtimes, long repair times, poor-quality products, unreliable suppliers, poor facility layout, inefficient material handling systems, deficient inventory control systems, large lot sizes, and long setup times.

Labor is wasted when the employee is assigned to produce unnecessary products, to move and store unnecessary inventory, to rework bad products, to wait during machine repairs, and to receive unnecessary training. Labor is also wasted when the employee makes mistakes due to lack of necessary training, individual responsibility, or mistake-proofing mechanisms.

From the JIT requirements point of view, all the subsystems in the manufacturing system must be improved.

8.6.1 JIT Impact on Facilities Design

There are many concepts and techniques related to the JIT production system that impact building design, facility layout, and the material handling system, such as

- Reduction of inventories
- Deliveries to points of use

- Quality at the source
- Better communication, line balancing, and multifunctional workers

8.6.1.1 Reduction of Inventories

One of the main objectives of the JIT production system is the reduction of inventories. Inventories can be reduced if products are produced, purchased, and delivered in small lots; if the production schedule is leveled appropriately; if quality control procedures are improved; if production, material handling, and transportation equipment are maintained adequately; if products are pulled when needed and in the quantities needed; and so on. Therefore, if inventories can be reduced,

1. *Space requirements are reduced*, justifying arranging the machines closer to each other and the construction of smaller buildings. Reducing the distances between machines reduces the handling requirements.
2. *Smaller loads are moved and stored*, justifying the use of material handling and storage equipment alternatives for smaller loads.
3. *Storage requirements are reduced*, justifying the use of smaller and simpler storage systems and the reduction of material handling equipment to support the storage facilities.

Consequently, the building could be smaller, a better plant layout could be used, fewer handling and storage requirements might be needed, and the material handling and storage equipment alternatives could move and store smaller loads.

8.6.1.2 Deliveries to Points of Use

If products are purchased and produced in smaller lots, they should be delivered to the points of use to avoid stockouts at the consuming processes. Products can be delivered to the points of use if the building has multiple receiving docks and if a decentralized storage policy is used. If products can be delivered to the points of use, the following scenarios could happen:

1. If parts are coming from many suppliers, several receiving docks surrounding the plant could be required. Receiving docks need extra space for parking of trucks plus equipment for unloading, doors, and so on. Additionally, depending on the number of loads arriving at each receiving dock, storage equipment will be needed at each decentralized storage area. Also, if parts are delivered to the points of use, then the building will need multiple receiving docks, storage and handling equipment will be needed at each receiving dock, and the parts could be moved shorter distances and fewer times (promoting a shorter production cycle and improving the inventory turnover, reducing the possibility of having bad quality due to excessive handling, and detecting bad-quality parts more quickly). The storage and handling equipment alternatives to support multiple receiving docks are usually simple and inexpensive (i.e., pallet racks, pallet jacks, pallet trucks, walkie stackers). Side-loading trailer trucks could also be used to avoid building expensive receiving docks.

2. A decentralized storage policy could be required to support the multiple receiving docks, and computerized or manual inventory control could be used. If inventory control is manual (using cards or kanbans), then a centralized kanban control system could be used to collect the kanbans and request more deliveries from the suppliers, or a decentralized kanban control system could be used with returnable containers and/or withdrawal kanbans being returned to the supplier after every delivery.
3. If plant layout rearrangements have been performed to support the JIT concepts, then internal deliveries to the points of use could be carried out using material handling equipment alternatives for smaller loads and short distances. If the JIT delivery concept is applied without performing layout rearrangements, then faster material handling equipment alternatives could be justified, depending on the "pulling rate" of the consuming processes.
4. If a truck is serving different receiving docks, it could be loaded based on the unloading sequence. Side-loading trucks could also be used. If the traffic around the plant is problematic, then a receiving terminal could be used to sequence the deliveries.

8.6.1.3 Quality at the Source

Every supplying process must regard the next consuming process as the ultimate customer and each consuming process must always be able to rely on receiving only good parts from its suppliers. Consequently, the transportation, material handling, and storage processes must deliver to the next process parts with the same level of quality that they received from the preceding process. To achieve the quality-at-the-source concept, the following could be required:

1. Proper packaging, stacking, and wrapping procedures for parts and boxes on pallets or containers
2. Efficient transportation, handling, and storage of parts
3. A production system, supported by teamwork, that allows the worker to perform his or her operation without time pressure

Therefore, if quality at the source is implemented, better packaging, stacking, and wrapping procedures could be required; transportation, storage, and handling must be carefully done; and asynchronous assembly systems, sometimes with U-configurations, could be used. The assembly systems could be supported by the team approach concept, and quick feedback procedures supported by electronic boards could be used to achieve line balancing.

8.6.1.4 Better Communication, Line Balancing, and Multifunctional Workers

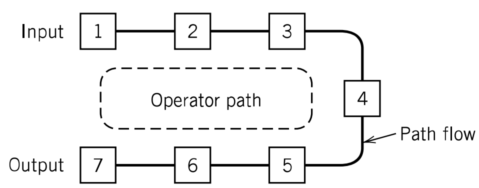
In many JIT manufacturing systems, U-shaped production lines are being used to promote better communication among workers, to use the multiple abilities of workers that allow them to perform different operations, and to easily balance to production line using visual aids and the team approach. Most of these applications have been very successful and have incorporated "problem boards" to write down the problems that occur during the shift, to analyze them at the end of the shift, and to generate solution methods to eliminate the future occurrence of these problems.

The U-shaped flow lines are discussed in the next section. Layout arrangements with U-shaped patterns minimize material handling requirements (in most cases, they eliminate them). They also support the pulling procedure with kanbans. In some companies in which this layout arrangement is in operation, inventory buffers between processes of only one part or one container are used.

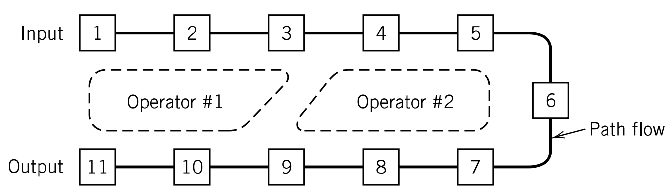
8.6.2 U-Shaped Flow Lines

A U-shaped flow line is a variation of the straight-line flow structure (Figure 8.7). It follows the concept of group technology, where parts of similar processing characteristics are grouped together for processing in a common area, or *manufacturing cell.* The shape is unique in that it allows more efficient use of operators tending the machines, and it promotes better communication among workers.

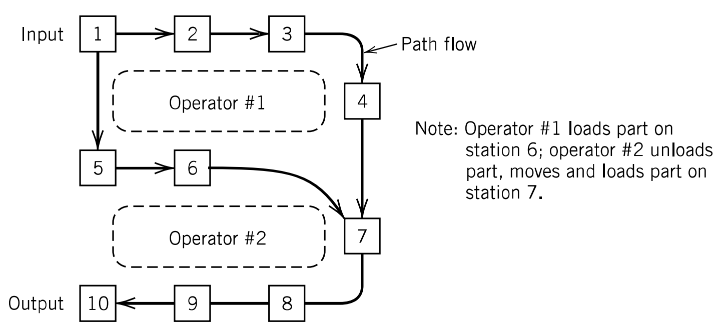
The transition from traditional processing to processing in a U-line has brought significant benefits to many companies. In a study of 114 U.S. and Japanese U-lines, Miltenburg [19] reports that "the average U-line has 10.2 machines and 3.4



(*a*) One-operator U-line



(*b*) Two-operator U-line



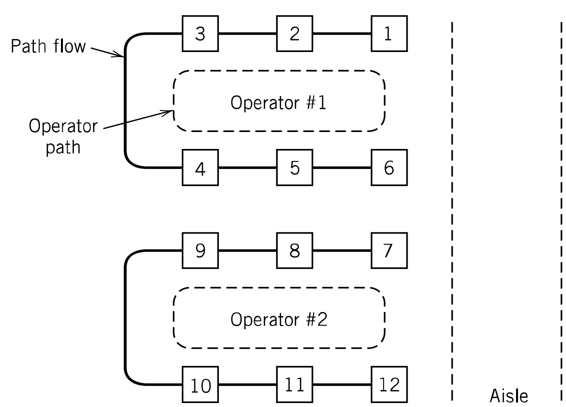
(*c*) Two-operator U-line with bypass

Figure 8.7 Illustrations of U-lines.

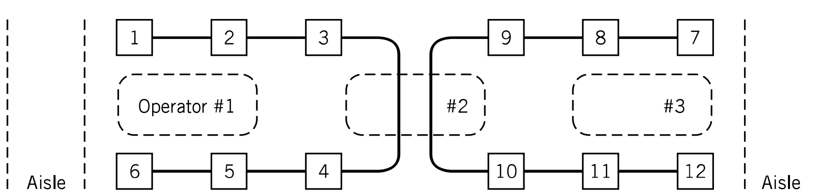
operators. About one-quarter of all U-lines are manned by one operator and so run in chase mode. The reported benefits are impressive. Productivity improved by an average of 76%. WIP dropped by 86%. Lead time shrank by 75%. Defective rates dropped by 83%."

The U-line comes in many different forms, depending on the number of products manufactured, the number of machines in the line, and the number of operators tending the machines. For a one-operator U-line, the machines are laid out in a U-pattern, and the operator works within the U-line. The machines are close together, and the space between is limited by the size of the machine. Since proximity is important in reducing the material handling time and the operator walking time between adjacent machines, there are benefits to using small and simple machines. There is also the goal of minimizing the inventory between machines to a maximum of one unit. The operator in this type of U-line is said to be in "chase mode"; that is, the operator is chasing the parts as they go from one machine to the next. Figure 8.7*a* illustrates a one-person U-line manufacturing or assembly cell. A two-operator U-line is illustrated in Figure 8.7*b*. A U-line with bypass is shown in Figure 8.7*c*. Other variations include the tandem-line, with separate U-lines on the same side of the aisle, and one where the second U-line is across the aisle from the first line. See Figure 8.8 for an illustration. The configuration shown in Figure 8.8*a* illustrates independent lines, while Figure 8.8*b* illustrates dependent lines. The dependency is based on using one of the operators assigned to tend machines on both lines.

More complex U-lines can be derived by using the simple U-line as a template. Miltenburg [19] shows a simple U-line with a bypass, embedded U-lines, a figure eight



(*a*) Independent U-lines



(*b*) Dependent U-lines

Figure 8.8 Independent vs. dependent U-lines.

pattern consisting of three lines with two operators, and a multi-U-line facility. The last category is designed to minimize the number of operators tending the machines.

One can undoubtedly come up with other types of U-line layouts based on the required number of machines and operators and the required output rate. The objective here is to find an efficient pattern that will minimize the time to move parts between machines and the time it takes the operator to complete a cycle.

U-lines are scalable since they are able to adjust the line output to conform to an increase or decrease in production requirements based on the demand rate (i.e., the line is flexible to production volume changes). Scaling is done by increasing or decreasing the number of operators tending the line. Increasing output on short notice requires that there be excess processing capacity in the line. Rebalancing of the line is done quite often to adjust the line to varying demand quantities. Rebalancing also involves adding or removing machines on the line or changing the standard times based on the new layout configuration. The operator walking time will change depending on the new layout of the line. Rebalancing involves determining the number of operators required and assigning the machines that each operator tends.

Flexibility is achieved by making machines movable so that the machines can be re–laid out to minimize part transportation time and operator walking time. Putting machines on standard platforms that are easily lifted and moved, or using casters, are two ways of making the machines movable. Quick setup is necessary so that new parts can are immediately processed. Flexibility also requires that operators perform multiple tasks and be knowledgeable and versatile enough to tend all machines in the cell. It also requires that the tasks to be performed are clearly defined and the operator is trained to perform the tasks properly.

Scalability and flexibility make U-lines one of the most desirable manufacturing cell configurations.

8.6.3 Remarks

There is another interpretation of what has been the main thrust of production systems such as transfer lines, flexible manufacturing systems, and JIT production systems. From a manufacturing point of view, productive time means that a product is being transformed from one state to another (e.g., metal blocks converted to machined parts, component parts assembled to form end products, etc.). The rest of the time is moving and waiting time. A more contemporary interpretation set within the context of lean manufacturing is the concept of value-adding and non-value-adding activities. Machining and assembly operations are value-adding activities, while transportation and storage are non-value-adding activities. Reduction in non-value-adding times will lead to shorter cycle times, and consequently, work-in-process inventories will be lower. Reduction in non-value-adding activities leads to lean manufacturing.

The lean manufacturing concept can be summarized as follows:

- Eliminate or minimize non-value-adding activities.
- Only produce what is demanded.
- Minimize the use of time and space resources.
- Manufacture in the shortest cycle time possible.

These concepts are consistent with the just-in-time manufacturing philosophy.

Other versions of the JIT manufacturing system include

- Stockless production
- Material as needed
- Continuous-flow manufacturing
- Zero-inventory production systems

We can expect to see the JIT manufacturing system continue to be refined by the requirements of individual companies. Additional information on JIT can be found in [20, 22, 25–30, 34].

8.7 FACILITIES PLANNING TRENDS

Factories are now being designed to provide the smoothest flow of materials, achieve flexibility (to adapt to proliferating product lines and volatile market demands), improve quality, increase productivity and space utilization, and simultaneously reduce facilities and operating costs. The following trends can be observed:

1. Buildings that have more than one receiving dock and that are smaller in size (less space requirements for staging, storage, warehousing, offices, and manufacturing primarily because of JIT purchasing, small lot production, deliveries to points of use, decentralized storage areas, better production and inventory control, visual management, manufacturing cells, Electronic Data Interchange (EDI), focused factories, open-layout offices, decentralized functions, and simplified organizational structures). Multiple receiving docks require efficient loading/unloading of side-loading trucks when used.
2. Smaller centralized storage areas and more decentralized storage areas (supermarkets) for smaller and lighter loads.
3. Decentralized material handling equipment alternatives at receiving docks.
4. Material handling equipment alternatives for smaller and lighter loads.
5. Visible, accessible, returnable, durable, collapsible, stackable, and readily transferable containers.
6. Asynchronous production lines with mixed-model sequencing for "star" products.
7. Group technology layout arrangements for medium-volume products with similar characteristics to support the cellular manufacturing concept.
8. U-shaped assembly lines and fabrication cells, and other variations of the U-pattern, as well as other flow patterns.
9. Better handling and transportation part protection.
10. Fewer material handling requirements at internal processes; more manual handling within manufacturing cells.
11. Standard containers, trays, or pallets.
12. Side-loading trucks for easier access and faster loading and unloading operations.
13. Changing the traditional function of storing to one of staging.

14. Accelerated use of bar codes, laser scanners, EDI, and machine vision to monitor and control the flow of units.
15. Radio Frequency Identification (RFID) technologies across the entire spectrum of manufacturing and service sectors, including manufacturing and healthcare industries.
16. Focused factories with product and/or cellular manufacturing and decentralized support efforts.
17. Lift trucks equipped with radio data terminals and mounted scales to permit on-board weighing.
18. Flow-through terminals and/or public warehouses to receive, sort, and route materials.
19. Facilities design process as a coordinated effort among many people.
20. Continued globalization of manufacturing activities in countries that are able to produce at lower costs.
21. Launching of bigger and more energy efficient mega-container ships that will drive down shipping costs between major international container ports.
22. Emergence of nano-sized products that will challenge contemporary thinking on concepts in manufacturing and supply chain systems.

8.8 SUMMARY

The objective of this chapter was to give a brief overview of manufacturing systems that may impact the facilities design function. Depending on the variety, volume, and value of products to be processed, different levels of automation, types of layout, and material handling systems will be appropriate. The design of transfer lines, flexible manufacturing systems, and single-step multimachine systems requires a variety of automated material handling devices. Design of flow systems for automated production systems brings us back to the fundamental issues addressed in the layout section of this book. The continued trend toward just-in-time manufacturing takes the material handling and layout functions to the front line. Are they necessary to begin with? Layout methods based on job shop assumptions may have to be forgone in favor of cellular-based layouts. The layout of U-lines is one that will continue to be of interest. And perhaps there will be other flow structures that can augment the U-line configuration.

Finally, as one might observe in recent publications, many products are getting smaller. For example, notebook computers are smaller and weigh less. Smaller and lighter products will continue to impact global manufacturing as international shipping costs will be lower. Additionally, the emergence of nanotechnology products and their conversion to industrial use will challenge current thinking on manufacturing methods. It should be obvious from our discussion that there is not a single method for designing the ideal manufacturing system. New materials are continually being introduced, and new manufacturing processes as well as methods of operation are being developed. Facilities planning tools are yet to be determined for these emerging products and processes. We believe, however, that the basic principles of facilities planning discussed in the book will continue to be applied in these new developments.